

Experimental Design Practical

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1. Human practical: Identification of prognostic biomarkers in human prostate cancer patients

A biomarker is any measurable biological characteristic that serves as an indicator of a specific disease state, reflecting its presence, progression, or severity. In oncology, biomarkers are invaluable clinical tools that can be non-invasively or minimally invasively quantified from a variety of biological matrices, including blood, urine, saliva, or solid tissue biopsies.

Specifically, prognostic biomarkers evaluate a patient's baseline risk, providing a statistical forecast of the likely course of the disease (such as overall or recurrence-free survival) independent of any specific therapeutic intervention.

Study aim:

The aim of this study is to identify a panel of prognostic biomarkers that can predict recurrence-free survival in patients with prostate cancer. Such biomarkers could help identify patients at higher risk of disease recurrence who may benefit from more aggressive treatment, while sparing patients with a more favourable prognosis from unnecessary treatment-related side effects and overtreatment.

1. What would the ideal study cohort be?
 - The ideal cohort would be a large, prospective, population-based cohort of all newly diagnosed prostate cancer patients in the UK. Patients would be enrolled at diagnosis, have biological samples collected before treatment, and be followed long-term to determine recurrence-free survival. The cohort would be representative of all ages, ethnicities, disease stages, and treatment pathways.
2. What factors make above ideal cohort difficult to achieve?
 - Several factors make ideal cohort difficult to achieve:
 - **Cost:** Recruiting and following a nationwide cohort is expensive.
 - **Time:** Prostate cancer recurrence may occur years after diagnosis, requiring long-term follow-up.
 - **Logistics:** Collecting, storing, and analysing samples from all patients is challenging.
 - **Patient consent:** Not all patients will agree to participate.
 - **Data completeness:** Clinical and follow-up data may be missing or inconsistent.
 - **Resource limitations:** Research staff, funding, and laboratory capacity are finite.
 - **Ethical considerations:** Ensuring patient privacy and data security on a large scale is complex.
3. In reality, what cohort would be chosen? What issues might this cause?
 - A more realistic cohort would consist of:
 - Patients recruited from a limited number of hospitals or cancer centres.
 - Individuals who consent to provide samples and clinical data.
 - Patients with available tissue samples and adequate follow-up information.
 - Potentially a retrospective cohort assembled from existing biobanks and medical records.
 - Using a more restricted cohort can lead to:
 - **Selection bias:** Participants may differ systematically from the wider prostate cancer population.
 - **Reduced generalizability:** Results may not apply to all patient groups, particularly underrepresented populations.
 - **Missing data:** Incomplete records can reduce statistical power and introduce bias.
 - **Confounding:** Differences in treatment, disease stage, or patient characteristics may influence recurrence-free survival independently of the biomarkers.

- **Limited event numbers:** Too few recurrence events may make it difficult to identify reliable prognostic biomarkers.
4. What are the pros and cons for using a retrospective or prospective cohort for this study? [A retrospective cohort study looks back over time and takes data from patients notes and/or interviews. A prospective cohort study follows the cohort through time collecting data as it goes].
- **Retrospective Cohort (Looking Backwards)**
 - Pros:
 - * **Immediacy (No Waiting for Events):** Because the 10-to-15-year clinical timeline has already happened, you instantly know which patients suffered a cancer recurrence and which did not. You can complete the study in months rather than decades.
 - * **Highly Cost-Effective:** You eliminate the massive costs associated with setting up a new clinical infrastructure, hiring long-term study coordinators, and tracking thousands of patients forward in time.
 - * **Leverages Existing Infrastructure:** The UK has world-class clinical biobanks and comprehensive centralized health registries (like the NHS) which make matching historical tissue samples to long-term clinical data highly feasible.
 - Cons:
 - * **Sample Degradation & Technical Noise:** Tissue blocks or blood samples stored for over a decade can experience chemical degradation. This introduces technical artifacts, where the biomarker results might reflect how poorly the sample was preserved over 10 years rather than the patient's actual tumor biology.
 - * **Treatment Confounding:** The standard of care for prostate cancer 10 to 15 years ago was different from today's medical guidelines. Changes in surgery techniques or older hormone therapies can mask or exaggerate the true prognostic power of your biomarker panel.
 - **Prospective Cohort (Moving Forwards)**
 - Pros:
 - * **High-Quality, Standardized Data:** By designing the study from the ground up, you can ensure that all samples are collected, processed, and stored using the same rigorous protocols. This minimizes technical variability and maximizes the reliability of your biomarker measurements.
 - * **Contemporary Relevance:** The prognostic biomarkers you identify will be directly applicable to current clinical practice, as they will be derived from patients treated under modern standards of care.
 - Cons:
 - * **Time-Intensive:** You will need to wait 10 to 15 years to observe enough recurrence events, which can delay the translation of your findings into clinical practice.
 - * **Costly:** The financial burden of recruiting and following a large cohort over many years is substantial, requiring sustained funding and resources.
 - * **Patient Attrition:** Over a long follow-up period, some patients may drop out of the study, move away, or be lost to follow-up, which can reduce the sample size and introduce bias if the attrition is not random.
5. What would the outcome measure(s) be for this study? What issues can arise in the definition of the outcome measures for this study?
- **Primary Outcome Measure**

- The primary outcome of this study is Recurrence-Free Survival (RFS). This time-to-event metric is defined as the duration (in months or years) from the date of primary definitive treatment to the date of documented prostate cancer recurrence or death from any cause, whichever occurs first.
 - To ensure statistical rigor, recurrence events will be strictly classified using standard clinical guidelines into one of the following mutually exclusive categories:
 - * **Biochemical Recurrence (BCR):** Defined by specific, sequential elevations in Prostate-Specific Antigen (PSA) thresholds (varying by primary treatment modality, such as post-prostatectomy vs. post-radiotherapy).
 - * **Radiological Recurrence:** Confirmed by imaging modalities (e.g., multiparametric MRI, PSMA PET-CT) showing new lesions consistent with prostate cancer.
 - **Secondary Outcome Measures**
 - **Distant Metastasis-Free Survival (DMFS):** Measured as the time from primary treatment to the first radiological confirmation of distant metastatic disease (e.g., bone or visceral lesions via bone scan or PSMA-PET).
 - **Overall Survival (OS):** Time from primary treatment to death from any cause, providing a holistic measure of patient prognosis.
 - **Issues in Outcome Definition:**
 - **Heterogeneity in Recurrence Definitions:** Different clinicians and institutions may apply varying criteria for defining biochemical recurrence, leading to inconsistencies in event classification across the cohort.
 - **Lead-Time Bias:** Patients with more intensive surveillance may have earlier detection of recurrence, artificially inflating RFS without truly improving patient outcomes.
 - **Competing Risks:** In an older population, non-cancer-related deaths can occur before a recurrence event, complicating the analysis of RFS and necessitating competing risk statistical models.
6. What factors might affect the outcome? What information may you need to collect for use in the study analysis?
- **Factors That Might Affect the Outcome (Confounders):**
 - **Tumor Aggressiveness:** High-grade tumors naturally recur faster regardless of biomarker levels.
 - **Treatment Differences:** Surgery, radiation, and hormone therapies (like Androgen Deprivation Therapy) have completely different baseline recurrence timelines.
 - **Surgical Precision:** Leaving positive surgical margins (cancer cells left behind) dramatically accelerates recurrence.
 - **Patient Health & Age:** Older patients with severe comorbidities (e.g., heart disease) often die of unrelated causes before their cancer has time to recur.
 - **Sample Quality:** The age of the archived sample and how it was processed can degrade the biological material, creating false biomarker readings.
 - **Information to Collect for the Analysis:** To isolate the true effect of your biomarkers using multivariate statistical models, you must collect the following baseline data for each patient:
 - **Clinical & Pathological Data:** Gleason Score/ISUP Grade Group, baseline PSA levels, and TNM staging.
 - **Treatment Details:** Type of primary treatment (surgery vs. radiation), surgical margin status (+ or -), and use of any secondary therapies (hormone therapy or chemotherapy).

- **Patient Demographics:** Age at diagnosis, ethnicity, and a comorbidity score (e.g., Charlson Comorbidity Index).
 - **Technical Quality Metrics:** Sample storage duration (years in the freezer) and an RNA/Protein integrity score to control for sample degradation.
7. What are your groups? Do you need a control group in addition? If so what will your control group be?
- After follow-up, patients could be divided into:
 - **Patients with recurrence (poor recurrence-free survival):** Prostate cancer recurs during the follow-up period.
 - **Patients without recurrence (good recurrence-free survival):** Remain recurrence-free throughout the follow-up period.
 - Do you need a control group in addition?
 - No, you do not need a separate control group. The 'control' in this study is the group of patients who do not experience recurrence. The comparison between the recurrent and non-recurrent groups will allow you to identify biomarkers that are prognostic of recurrence risk.
8. What effect would a short follow-up time have? Which group will this affect more?
- Effect of a short follow-up time:
 - Fewer recurrence events would be observed, reducing the statistical power of the study.
 - Some patients who will eventually develop recurrence may be incorrectly classified as recurrence-free.
 - The prognostic value of biomarkers may be underestimated because late recurrences are missed.
 - Survival estimates may be biased and less reliable.
 - It becomes more difficult to distinguish between patients with genuinely good prognosis and those who simply have not been followed long enough.
 - The non-recurrence (good prognosis) group is affected the most. This is because some patients classified as 'recurrence-free' at the end of a short follow-up period may actually experience recurrence later. These patients are therefore misclassified.
9. Which technologies could you use to screen for your panel of biomarkers?
- Genomics: RNA-seq, qPCR/RT-PCR
 - Proteomics: mass spectrometry, tissue microarrays
10. What factors might influence your choice of technology for this study?
- The decision would be based on your objectives, cost, logistics, feasibility and technological limitations.
11. Once your study has a panel of biomarkers from which a test for recurrence has been formulated, what is the next step?
- The test must be applied to an external validation cohort. This would be a different set of patients, perhaps in a different hospital and at a different time. The test will measure how well it predicts recurrence in prostate cancer.
12. What characteristics would a clinical test formed from your biomarkers need?
- **Sample Accessibility and Clinical Utility::** The test should provide rapid, on-site results during a clinic visit, eliminating laboratory processing delays.
 - **Non-Invasive Matrices::** Sampling should utilize non-invasive fluids like urine or saliva, rather than painful, clinically complex needle tissue biopsies.

- **Target Analytes:** Focus should be placed on secretory or soluble proteins that are actively shed into bodily fluids for easy detection via liquid biopsy.
- **Prioritizing High Specificity:** Because a positive result triggers aggressive secondary treatments, maximizing specificity is critical. This minimizes false positives, sparing low-risk patients from the harmful side effects of unnecessary over-treatment.

2. Mouse practical (Hypothetical example): Elucidating the Transcriptomic Landscape and Signaling Pathways Regulated by Hyperactive HHEX During Murine Organogenesis

The homeobox transcription factor HHEX is a critical orchestrator of murine brain and liver development. While wild-type HHEX undergoes precise regulatory control, a mutant variant—characterized by the substitution of all Serine and Tyrosine residues with Alanine—exhibits constitutive hyperactivity, resulting in embryonic lethality in homozygous mice. This study aims to delineate the downstream molecular mechanisms and aberrant gene regulatory networks driven by HHEX hyperactivity. Using embryonic day 15 (E15) fetuses, we will isolate brain and liver tissues across three distinct morphological genotypes: homozygous wild-type ($HHEX^{Wt/Wt}$), heterozygous mutant ($HHEX^{Wt/Mt}$), and homozygous mutant ($HHEX^{Mt/Mt}$). Through comparative transcriptomic profiling, we seek to identify global gene expression alterations and map the key signaling pathways disrupted by hyperactive HHEX, providing deeper insights into its role as a master developmental regulator.

Experimental design related question and answers:

- 1) What are the scientific questions of interest in this experiment?
 - Which genes show expression changes in response to mutant HHEX in brain and/or liver?
 - Do homozygous and heterozygous mutant HHEX mice show similar expression profiles?
 - Which genes show exclusive response in brain and liver respectively in response to hyperactivity of HHEX gene?
 - Which are the pathways that are involved in HHEX hyperactivity?
- 2) What are the independent and dependent variables in this experiment?
 - The independent variables are the factors that you, the investigator, choose to vary or group your subjects by. In this study, you have two independent variables, making this a factorial design:
 - Genotype (3 levels):
 - * Homozygous Wildtype (\$Wt/Wt\$)
 - * Heterozygous Mutant (\$Wt/Mt\$)
 - * Homozygous Mutant (\$Mt/Mt\$)
 - Tissue/Organ Type (2 levels)
 - * Brain
 - * Liver
 - The dependent variables are the outcomes you measure in response to the changes in the independent variables. In a genomics experiment, you are capturing high-dimensional molecular readouts:
 - Gene Expression Levels: The specific abundance of mRNA transcripts for every measured gene (quantified via methods like RNA-seq or microarray).
- 3) What are the null and alternative hypotheses in this case?
 - The Null Hypothesis (\$H_0\$): The null hypothesis is the position of 'no effect' or 'no difference'. It assumes that the hyperactive mutation in the HHEX transcription factor does not alter the regulation of the gene being tested.
 - In plain terms: The average expression level of Gene \$X\$ is exactly the same across all three groups of fetuses. The hyperactivity of HHEX has no molecular impact on this gene in that specific tissue.

- The Alternative Hypothesis (H_a): The alternative hypothesis is what you hope to discover.
 - In plain terms: Hyperactive HHEX mutation does cause a significant shift in gene expression.
- 4) What control samples should be included in this experiment, and why are controls necessary?
- Biological Controls: Homozygous Wildtype (Wt/Wt) Fetuses (True Negative Control)
 - These are your primary control samples. They contain normal, un-mutated HHEX transcription factors and display healthy morphology. Everything you discover about the hyperactive mutants (Mt/Mt) will be measured as a direct change relative to this baseline group.
 - Why is a control needed in this experiment?
 - Without controls, high-throughput genomics experiments like RNA-sequencing yield uninterpretable data. Controls are mandatory for two major reasons:
 - * Reason A: To Establish a Developmental 'Baseline':
 - Hundreds of genes are naturally turning on and off in the brain and liver as these organs rapidly mature.
 - If you only look at the gene expression of the mutant (Mt/Mt), you cannot tell whether a highly active pathway is a normal part of being a 15-day-old fetus, or if it was caused by the hyperactive HHEX mutation. The Wt/Wt group establishes what 'normal development' looks like at day 15.
 - * Reason B: To Account for Technical Noise
 - Isolating RNA from tiny fetal organs, preparing sequencing libraries, and running genomic chips introduces massive amounts of technical variability. Minor fluctuations in incubator temperature, pipetting precision, or a change in the batch of chemical reagents can alter your gene expression readings.
 - By running your Wt/Wt control samples in the exact same batch alongside your mutant samples, you can subtract this background technical noise, ensuring that the remaining differences are strictly biological.
- 5) How many replicates you need to include for each group? Discuss what factors might have influence in selecting the number of replicates?
- For a rigorous genomics experiment (like RNA-seq) looking at embryonic tissue development, a standard rule of thumb is to include a minimum of $n = 3$ to $n = 5$ biological replicates per group (meaning 3 to 5 completely independent fetuses per genotype, per tissue).
 - However, there is no single 'magic number.' Choosing the ideal number of replicates requires balancing several biological, statistical, and practical factors.
 - Factors Influencing the Number of Replicates: Selecting your sample size (n) is fundamentally an exercise in maximizing statistical power (the probability of detecting a true gene expression change if one exists). Some of the factors determine how many replicates you need:
 - Expected Biological Variability (Variance)
 - * Higher variability broadens the hypothesis distribution curves, increasing their overlap and reducing power. To counteract high variability and shrink your standard error, you must increase the number of biological replicates.
 - * Healthy Groups: In your healthy wildtype (Wt/Wt) and heterozygous (Wt/Mt) groups, biological variation might be relatively low, especially if the mice share a highly uniform genetic background.
 - * The Mutant Group: Your homozygous mutants (Mt/Mt) manifest severe morphological defects and undergo fetal death. Because different fetuses will be at slightly different stages

of developmental arrest or tissue degradation, the biological variability within the \$Mt/Mt\$ group might be high.

- Expected Effect Size: The effect size is the magnitude of the difference in gene expression between your groups.

- * Your mutant HHEX transcription factor is hyperactive, meaning it is expected to aggressively over-activate or repress its target downstream genes. This may translates to a very large biological effect size.
- * On a hypothesis chart, a large effect size physically pushes the alternative distribution far away from the null distribution, minimizing their overlap. Because the signal is expected to be so strong, you can achieve high statistical power to detect the primary target pathways with a relatively modest number of replicates (\$n = 3\$ or \$n = 4\$).
- * Note: If you also want to detect subtle, minor secondary gene modifications in the heterozygous (\$Wt/Mt\$) mice, you will need more replicates to capture those smaller effect sizes.

- The Specific Tissue Being Analyzed

- * Liver tissue at embryonic day 15 is relatively uniform, so lower variability is expected. Therefore, fewer replicates are likely sufficient.
- * Brain tissue, however, is incredibly heterogeneous, containing a complex mix of neural progenitors, migrating neurons, and glial cells. Because a physical brain dissection might capture slightly different ratios of these distinct cell types from mouse to mouse, the brain samples will inherently possess more 'cellular noise' than liver samples. A noisier tissue requires a higher replicate count to achieve the same level of statistical precision.

6) Which experimental groups will be included in the study?

- There would be six experimental groups:
 - 1) Brain from homozygous wild type mice
 - 2) Brain from homozygous mutant mice
 - 3) Brain from heterozygous mutant mice
 - 4) Liver from homozygous wild type mice
 - 5) Liver from homozygous mutant mice
 - 6) Liver from heterozygous mutant mice.
- This is a two-factor experiment with three levels and two levels, respectively.

	Wt.Wt	Wt.Mt	Mt.Mt
Brain	4	4	4
Liver	4	4	4

- Wt = Wild type Mt = Mutant

7) Assuming four replicates per group, calculate the total number of mice needed for this experiment. Consider two possible approaches: collecting both brain and liver tissue from the same animal, or using dedicated mice for each tissue type. Which approach would you recommend, and what is your reasoning?

- Calculation of Mice Needed

- Approach 1: Collecting both brain and liver tissue from the same animal

- * Total of six groups with three genotypes
- * For each genotype collect liver and brain tissue from same mouse
- * total mice required 12 (3 x 4)
- Approach 2: Using dedicated mice for each tissue type
 - * Total of six groups
 - * Replicates for group is 4
 - * Calculation: total of 24 mice required (6 x 4)
- Approach 1 (collecting both tissues from the same animal) is highly recommended because of the following reasons.
 - **The Principle of Reduction (Ethical Consideration):** In animal research, adhering to the 3Rs (Replacement, Reduction, Refinement) is mandatory for institutional approval (IACUC). Approach 1 cuts the number of animals required for the final experiment exactly in half (12 vs. 24). Choosing an approach that sacrifices 12 additional fetuses without a strict scientific necessity would violate ethical standards.
 - **Power of Intra-Individual (Paired) Analysis:** By extracting both tissues from the same mouse, you gain a massive analytical advantage: you can pair the data. Because you expect high variability in the \$Mt/Mt\$ group due to varying stages of fetal decline, you can directly correlate the severity of pathway disruption in the brain with the severity of pathway disruption in the liver within the exact same biological system. Approach 2 completely destroys this link, treating the brain and liver as unrelated entities.

8) How will any findings be validated?

- In high-throughput genomics experiments (like RNA-seq or microarrays), your initial dataset will identify hundreds of Differentially Expressed Genes (DEGs) and candidate pathways. However, due to the sheer volume of data and inherent technical noise, these findings are considered predictive correlation.
- To transform these data points into a scientifically accepted discovery, you must validate your findings using a combination of independent experimental techniques and orthogonal biological approaches like below.
 - Quantitative Reverse Transcription PCR (RT-qPCR)
 - Western Blotting or ELISA (Protein-Level Validation)

9) What are possible sources of bias and confounding variables in the experiment?

- Some possible sources of bias:
 - **Observer / Confirmation Bias (The Visibility Confound):**
 - * **Source:** This experiment explicitly states that the \$Wt/Wt\$, \$Wt/Mt\$, and \$Mt/Mt\$ day-15 fetuses manifest distinguishing morphological characteristics. When you open the uterus to harvest the organs, you will immediately know which embryos are the mutants (\$Mt/Mt\$) just by looking at them. This can subconsciously influence how carefully or quickly you dissect the liver or brain, leading to variations in tissue quality or RNA stability.
 - * **Mitigation:** Implement strict blinding. Have one lab member harvest the fetuses from the mother and assign a randomized numerical code to each embryo. A second researcher—who has no idea which code belongs to which phenotype—should perform the micro-dissections and RNA extractions.
 - **Technical Selection Bias:**

- * **Source:** Because the hyperactive HHEX mutation induces fetal death, some of the \$Mt/Mt\$ embryos harvested at day 15 may have already stopped developing or died hours before the dissection, while others might still be alive but severely malformed. If you accidentally select only the 'healthiest-looking' mutants to achieve technical consistency, you introduce selection bias and miss the true, catastrophic gene expression changes that drive the lethality.
- * **Mitigation:** Establish strict inclusion/exclusion criteria prior to the experiment. For example, measure the Crown-Rump Length (CRL) of the fetuses, and use an automated assay (like an RNA Integrity Number / RIN score) to verify that the tissue has not undergone necrotic post-mortem degradation before passing it to sequencing.
- Some confounding Variables:
 - **The Litter Effect:**
 - * **Source:** To get all your samples, you will need to breed multiple pregnant female mice. Different mothers will have slight variations in their personal metabolism, stress levels, cage position on the facility racks, and even how many total pups they are carrying in their uterus (litter size). If you accidentally harvest your \$Wt/Wt\$ controls from Mother A and your \$Mt/Mt\$ mutants from Mother B, the 'Litter Effect' will completely confound your data. Any gene expression differences you find might just be due to Mother A's diet versus Mother B's stress levels.
 - * **Mitigation:** Use Randomized Block Design. Treat each individual pregnant mother (litter) as a 'block.' Ensure that your 4 replicates for each genotype are distributed evenly across multiple litters.
 - **Dissection Time and Technical Drifts (Batch Effects):**
 - * **Source:** Processing 12 to 24 micro-dissections takes time. If you dissect all the wildtype brains first, and by the time you reach the final mutant livers two hours have passed, the samples processed later will suffer from transcriptional drift (the tissue reacting to being out of the body) or slight RNA degradation.
 - * **Mitigation:** Alternating or randomizing the order of dissection is mandatory. Do not process your groups sequentially by genotype. Instead, flip a coin or use a random number generator to determine the order of dissection (e.g., Mutant 1, then Wildtype 1, then Heterozygote 1).
 - * Other possible confounders:
 - * Age of the mice, animal breed, diet etc
 - * People, if several people involved in sample collection and extraction.
 - * Sequencing run
 - * Cage position
 - * Technician handling the mice
 - * Extraction batch
 - * sex
 - * ...

10) How can these sources of bias and confounding be controlled?

- Remember the principle: 'Block what you can; randomize what you cannot.'